

Electric-Field-Switchable Altermagnetism *via* Ligand Rotation in a d^0 Metal–Organic Framework

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Abstract

Altermagnetism is an emerging collinear magnetic phase that combines vanishing net magnetization with momentum-dependent spin splitting. While altermagnetic signatures have recently been identified in metal–organic frameworks (MOFs), and a spin-crossover route to switching has been proposed, a non-invasive and reversible external control remains to be demonstrated. Here we report an electric-field-switchable route to altermagnetism in a flexible d^0 -MOF, where rotation of the organic ligands acts as the switching degree of freedom. In the planar ground state, active p_z orbitals mediate dominant superexchange through the 180° ligand–metal–ligand pathways (J_2), stabilizing a conventional antiferromagnetic state. Rotating the ligands out of plane activates the in-plane p_x/p_y orbitals, strongly enhances the 90° pathways (J_1), and reverses the J_1/J_2 competition. The antiferromagnetic-to-altermagnetic crossover sets in at a moderate tilt angle of $\approx 34^\circ$, reaching a spin splitting of 84 meV at the fully rotated 90° configuration. The 90° altermagnetic state can be switched on by the electric field and switched off upon field removal, as the constrained-rotation landscape decreases monotonically back toward 0° , enabling barrierless relaxation to the antiferromagnetic state and reversible on/off operation. Ab initio molecular dynamics at 300 K shows that structural thermal fluctuations transiently rotate individual ligands across the threshold but without spatial coherence across the lattice, underscoring the essential role of the electric field in synchronizing the collective rotation. With polar substituents such as fluorination, this locking is achieved at a

critical field of ≈ 0.22 V/Å, comparable to fields applied in field-effect devices and below typical dielectric breakdown limits. Our work establishes a paradigm in which altermagnetic order emerges from ligand *p* orbitals rather than transition-metal *d* orbitals, and structural flexibility enables electric-field control, opening an avenue for responsive organic spintronics.

1. Introduction

Altermagnetism has recently been established as a new class of collinear magnetic materials that complements the conventional dichotomy of ferromagnetism and antiferromagnetism.¹⁻⁶ Its defining feature is a symmetry-driven arrangement in which opposite spin sublattices are connected by rotational operations, rather than by inversion or translation, enabling nonrelativistic spin splitting in the electronic band structure despite vanishing net magnetization.^{1,2} This unique combination endows altermagnets with time-reversal-symmetry-breaking magnetoelectric responses akin to those of ferromagnets, while preserving the advantages of antiferromagnetic order, zero stray fields, robustness against external magnetic perturbations, and terahertz-scale spin dynamics.^{7,8} The spin-split bands also allow electrical readout of the magnetic state. These attributes make altermagnets particularly attractive for information storage and spin-based logic devices.^{7,8}

Recently, a wide range of altermagnetic systems have been experimentally confirmed or theoretically predicted, spanning inorganic compounds such as rutile-type RuO₂, hexagonal semiconductors MnTe and CrSb, and more recently metal-organic frameworks (MOFs) that have attracted considerable attention.⁹⁻¹⁵ The vast majority of altermagnetic systems reported to date follow a static design paradigm: the spin splitting is dictated by fixed lattice symmetries and, once synthesized, remains locked in place, producing only a single fixed magnetic state. External-stimulus control has begun to emerge in inorganic and van der Waals platforms, including ferroelectric-field manipulation,¹⁶ twist-angle engineering,¹⁷ bilayer stacking design,¹⁸ strain,¹⁹ and charge-order driving²⁰. Nevertheless, a mechanism that enables reversible switching between two distinct magnetic states, such as the 0 and 1 required in memory or logic

operations, remains highly desirable, particularly in chemically tunable material platforms.

Metal-organic frameworks (MOFs) are crystalline porous materials constructed from metal nodes and organic linkers through coordination bonds.²¹⁻²³ Their chemical modularity offers unique advantages for designing magnetic materials.²⁴⁻²⁷ By varying the metal centers,^{28,29} ligand types,^{30,31} and their connectivity,^{32,33} the lattice topology, electronic structure, and magnetic interactions of MOFs can be systematically tuned. More importantly, unlike rigid inorganic crystals, the organic linkers in MOFs are not static structural units. They can undergo conformational changes such as rotation and twisting.^{34,35} This inherent structural flexibility provides a physical basis for responding to external stimuli. Recent theoretical work has demonstrated that MOFs can host altermagnetic states.³⁶⁻⁴³ Che and co-workers designed organic ligands with nonbonding molecular orbitals to realize d-wave altermagnetic spin splitting in Cr-based 2D MOFs.³⁶ Baldoví and co-workers exploited ligand symmetry to lower the lattice symmetry, achieving g-wave and d-wave altermagnetism in imidazole-based 2D MOFs.⁴⁰ These studies confirm that the chemical tunability of MOFs offers a broad design space for altermagnetic materials. However, these existing MOF-AM works still follow the static design logic. They "write" the altermagnetic state into the lattice by selecting specific ligands and metal nodes, leaving the magnetic state fixed once synthesized.⁴¹⁻⁴³ External control of altermagnetism in MOFs has so far been limited to tuning the spin-polarized currents within a fixed altermagnetic state.⁴² Only very recently have two switching strategies been proposed: ferroelectric switching of the altermagnetic spin splitting in a hybrid perovskite MOF, where the control is mediated by the ferroelectric polarization rather than acting directly on the magnetic lattice,⁴⁴ and pressure-induced spin-crossover switching in a layered van der Waals MOF, which operates through a spin-state transition localized on the metal centers.⁴⁵ Yet a clean, non-invasive, and reversible route to direct electric-field switching between the antiferromagnetic and altermagnetic states remains unexplored.

Here we propose an approach that exploits the conformational flexibility of ligands as a tuning degree of freedom to achieve reversible switching of the altermagnetic state

in MOFs. To clearly demonstrate this mechanism, we choose a d^0 -MOF as our model system. The d^0 metal centers carry no magnetic moment, so the magnetic response originates entirely from the p electrons of the ligands. This eliminates the complexity of metal d electrons and allows us to directly track how ligand conformational changes affect the magnetism. In this system, the in-plane (0°) and out-of-plane (90°) configurations of the ligands correspond to different active orbitals, p_z versus p_x/p_y . This orbital switching directly alters the relative strength of two superexchange pathways through the metal centers. The 180° pathway (J_2) dominates at 0° , stabilizing a conventional antiferromagnetic ground state, while the 90° pathway (J_1) is significantly enhanced upon ligand rotation. The J_1/J_2 competition reverses at approximately 34° , driving the system into the altermagnetic state. At the fully rotated 90° configuration, the system achieves an 84 meV spin splitting, comparable to the largest ligand-engineered values in MOF altermagnets,⁴⁰ yet arising entirely from ligand p orbitals rather than metal d states. The 90° configuration, however, is energetically higher than the 0° configuration and is metastable. Thermal fluctuations can only transiently populate this state without establishing long-range order. This necessitates an external locking mechanism. By introducing polar substituents on the ligands, we demonstrate that an external electric field of approximately $0.22 \text{ V/\AA}$ is sufficient to synchronously lock all ligands into the 90° configuration, thereby establishing long-range altermagnetic order. Upon removing the field, the constrained-rotation energy landscape decreases monotonically back to the planar ground state, leading to barrierless relaxation to the 0° antiferromagnetic state and completing a reversible switching cycle. Our work establishes a new paradigm of p -electron altermagnetism, advancing the control of altermagnetic states from static materials design to dynamic manipulation of ligand conformation. The combination of MOF structural flexibility, ligand p -electron magnetism, and electric-field reversible switching opens a new avenue for organic spintronics.

2. Results and discussion

2.1 Design concept

The design strategy of our 2D d^0 -MOF platform is illustrated in Figure 1. The framework adopts a square-lattice topology with d^0 metal centers coordinated by organic ligands, yielding a metal-to-ligand ratio of 1:2. The metal centers are nonmagnetic, ensuring that all magnetic responses originate solely from the ligand sublattice. This clean electronic structure allows us to isolate the role of ligand geometry in determining the magnetic ground state. **Figure 1a** schematically illustrates the two limiting ligand configurations: the planar configuration (0°) on the left and the perpendicular configuration (90°) on the right, with the ligand π -orbitals explicitly depicted. In the planar configuration, the ligand π -orbitals are oriented out of the plane. Two distinct superexchange pathways are identified: the nearest-neighbor pathway J_1 (90° ligand-metal-ligand) along the sides of the square, and the next-nearest-neighbor pathway J_2 (180° ligand-metal-ligand) along the diagonals. The out-of-plane π -orbitals are more effectively coupled through the metal centers along the diagonal 180° pathway (J_2) than along the sides (J_1), making J_2 the anticipated dominant exchange channel. This J_2 -dominated exchange is expected to stabilize an antiferromagnetic (AFM) arrangement, with the magnetic moments on the four ligands adopting an $\uparrow\uparrow\downarrow\downarrow$ spin configuration. In the perpendicular configuration, the π -orbitals reorient to lie in the plane. This reorientation enhances the coupling through the metal centers along the 90° pathway (J_1) while suppressing the 180° pathway (J_2), reversing the J_1/J_2 competition. The resulting J_1 -dominated exchange is expected to stabilize an altermagnetic (AM) state, with the magnetic moments adopting an alternating $\uparrow\downarrow\uparrow\downarrow$ arrangement. The bidirectional arrow between the two configurations indicates the orbital-driven J_1/J_2 switching that underlies the magnetic-state transition. **Figures 1b** and **1c** schematically show the corresponding band structures. In the AFM case (Figure 1b), the spin-up and spin-down bands remain degenerate throughout the Brillouin zone. In the AM case (Figure 1c), the degeneracy is lifted, producing momentum-dependent spin splitting, a hallmark of altermagnetism.

2.2 Structural Model and Magnetic State Analysis

The 2D MOF monolayer is constructed using Ca centers as the metal nodes and

terephthalonitrile (TPN, $C_8N_2H_4$) as the organic linkers. Each TPN ligand coordinates to the Ca centers through its two nitrile nitrogen atoms. The Ca center adopts a square-planar coordination geometry with four N atoms from four distinct TPN ligands, yielding a metal-to-ligand ratio of 1:2. The resulting monolayer features a tetragonal lattice with C_4 rotational symmetry and belongs to the $P4/mmm$ space group. Owing to the d^0 electronic configuration of Ca, the metal centers carry no local magnetic moment. Consequently, all magnetic responses originate entirely from the ligand sublattice, providing a clean platform to investigate the role of ligand geometry in determining the magnetic ground state. The flexible metal-ligand coordination bonds allow the TPN ligands to rotate around the Ca-N bond axis, giving rise to two limiting conformations: the planar configuration (0° , **Figure 2a**), where the ligand molecular plane lies parallel to the xy plane, and the perpendicular configuration (90° , **Figure 2b**), where the ligand plane is oriented perpendicular to the xy plane. To properly describe the magnetic structure in the 90° configuration, a $\sqrt{2} \times \sqrt{2}$ supercell is adopted. The optimized lattice parameters are summarized in **Table S1**.

To determine the magnetic ground states of the two limiting configurations, we compared the total energies of four magnetic phases, antiferromagnetic (AFM), altermagnetic (AM), ferromagnetic (FM), and nonmagnetic (NM), for both the 0° and 90° ligand conformations (**Figure S1** and **Table S2**). The AFM state is the lowest in energy for the planar configuration (0°), while the AM state is energetically favored for the perpendicular configuration (90°). Across the two conformations, the planar configuration is energetically favored by 0.36 eV, identifying the 0° AFM state as the global ground state and the 90° AM state as metastable. Compensating this energetic cost requires an external driving force; as we show in Section 2.5, an electric field acting on appropriately polarized ligands provides such a force.

We then calculated the electronic band structures for both configurations. For the 0° planar configuration (**Figure 2c**), the spin-up and spin-down bands remain fully degenerate throughout the Brillouin zone, consistent with the AFM state. In contrast, the 90° perpendicular configuration (**Figure 2d**) exhibits clear spin splitting along the

M- Γ -M' high-symmetry path, with a maximum splitting of 84 meV at the valence band maximum. This momentum-dependent spin splitting is a hallmark of altermagnetism, confirming the AM nature of the 90° configuration.

The projected density of states (PDOS) analysis (**Figure 2c** and **2d**, right panels) shows that the electronic states near the Fermi level in both configurations are predominantly contributed by the ligand orbitals, with negligible contribution from the Ca centers. This confirms that the magnetic and electronic properties are governed entirely by the ligand sublattice. The 84 meV splitting is comparable to the largest values achieved in ligand-engineered Cr-based 2D MOF altermagnets (83.9 meV in the *d*-wave and 65 meV in the *g*-wave cases⁴⁰), and its origin from ligand orbitals rather than metal *d* states highlights the *p*-electron character of the altermagnetism in this *d*⁰-MOF platform.

To understand the evolution of the magnetic ground states with ligand rotation, we calculated the relative energies of the AFM, AM, and FM states as a function of the ligand tilt angle (**Figure 2e**). The AFM state remains lowest in energy below ~34°, while the AM state becomes favored above this angle, identifying the critical tilt angle for the AFM-AM transition. To trace the origin of this transition, we extracted the exchange parameters J_1 and J_2 by mapping the total energies of the magnetic configurations onto a classical Heisenberg model (**Figure 2f**). At 0°, J_2 dominates over J_1 , stabilizing the AFM ground state. As the ligand tilts, J_1 increases while J_2 is suppressed. The two exchange pathways cross at ~34°, after which J_1 dominates and stabilizes the AM state. The angular dependence of J_1 and J_2 thus directly correlates with the magnetic phase transition, confirming the design concept that ligand geometry controls the magnetic ground state through the competition between J_1 and J_2 .

2.3 Orbital Mechanism

The microscopic origin of the J_1/J_2 reversal is elucidated by examining the spin-resolved charge density and the orbital-resolved density of states for both configurations (**Figure 3**). In the planar configuration (0°, **Figure 3a** and **3b**), the spin charge density is uniformly dispersed over the TPN ligand framework, with spin-

polarized electrons predominantly occupying the out-of-plane p_z orbitals of the N and 1,4-site C atoms (**Figure S2**). The electronic states near the Fermi level are dominated by p_z orbitals, while the contributions from in-plane p_x/p_y orbitals are negligible. This out-of-plane orbital distribution enables efficient coupling along the diagonal 180° pathway (J_2) through the metal centers, strengthening the J_2 -mediated exchange and stabilizing the AFM state. In the perpendicular configuration (90° , **Figure 3c** and **3d**), the ligand rotation reorients the π orbitals to lie in the plane, exposing the p_x/p_y orbitals that now dominate the electronic states near the Fermi level. This orbital occupation reconstruction enhances the orbital overlap along the 90° pathway (J_1) through the metal centers, while simultaneously suppressing the 180° pathway (J_2). Consequently, J_1 dominates and the AM state is stabilized.

The orbital switching from p_z to p_x/p_y upon ligand rotation thus provides the microscopic mechanism for the J_1/J_2 reversal. The spatial reorientation of the ligand π orbitals selectively modulates the two superexchange pathways through the metal centers, directly controlling the magnetic ground state.

2.4 Thermal Fluctuations

The preceding sections have established that ligand rotation can switch the magnetic ground state from AFM to AM at the level of static DFT calculations. However, whether this switching can be realized under finite-temperature conditions remains to be verified. We therefore performed ab initio molecular dynamics (AIMD) simulations at 300 K to assess the structural stability and the thermal behavior of ligand orientations.

The total energy trajectory in **Figure 4a** shows stable oscillations around a constant value over 10 ps, confirming that the 2D framework maintains its structural integrity at room temperature. Despite this stability, thermal fluctuations are sufficient to activate ligand rotation. We sampled 50 snapshots evenly distributed over the trajectory and classified them according to the number of ligands exhibiting large-angle tilts exceeding the critical angle of $\sim 34^\circ$ (**Figure 4b**). The statistics reveal that single-ligand flip events account for nearly 50% of the sampled snapshots, and dual-ligand flips appear with a similar probability. Three-ligand tilts occur only rarely, while

simultaneous four-ligand tilts are not observed within the simulation time scale.

We then extracted the average lifetimes of these thermally activated configurations (**Figure 4c**). Single-ligand flips persist for approximately 600 fs, while dual-ligand flips have an even shorter lifetime of about 400 fs. These values are far too short to establish long-range cooperative order. Moreover, the electronic band structures of the transient configurations (**Figure S3**) show that the spin splitting is substantially reduced compared to the fully ordered 90° AM state (only ~15 meV for the dual-flip case), confirming that thermal fluctuations alone cannot establish a robust AM character.

These results demonstrate a clear limitation: although thermal fluctuations can transiently drive individual ligands across the switching threshold, the resulting configurations are short-lived, lack spatial coherence, and cannot sustain the long-range orientational order required for the AM state. An external control mechanism is therefore essential to synchronize and stabilize the ligand orientations. This motivates the electric-field approach presented in the following section.

2.5 Electric-Field Switching of the Altermagnetic Phase

The preceding sections have shown that ligand rotation can switch the system between AFM and AM, and that thermal fluctuations alone cannot stabilize the AM state because they only produce transient, incoherent local tilts without long-range orientational order. An external control mechanism is therefore required to synchronize and lock the ligand orientations. Electric-field control offers a natural route to achieve this,^{46,47} but the pristine TPN ligand is centrosymmetric and lacks a molecular dipole moment, rendering it unresponsive to electric fields. To enable electric-field coupling, we introduce a half-fluorinated TPN ligand (**Figure 5a**), which breaks the inversion symmetry and creates a net dipole moment in the 2D MOF (**Figure 5b**).

We first verified that fluorination does not alter the intrinsic magnetic behavior of the system. The energy evolution, exchange competition, and band structures of the F-system are all consistent with those of the H-system (**Figures S4-S6**), confirming that the 0° AFM and 90° AM ground states, as well as the J_1/J_2 switching mechanism, remain unchanged upon fluorination. Similar behavior is also observed for methyl

substitution (**Figures S7-S8**), confirming that the strategy is not limited to fluorination.

We then examined the response of the F-system to an external electric field applied along the out-of-plane direction. At zero field, the 0° AFM configuration is the ground state, while the 90° AM configuration is metastable. When an electric field is applied, the dipole moment of the fluorinated ligand couples to the field, lowering the energy of the 90° configuration relative to the 0° configuration. We calculated the energy difference between the two configurations as a function of field strength (**Figure 5c**). At $0.2 \text{ V/\AA}$, the 0° configuration remains lower in energy. The two configurations become nearly degenerate at approximately $0.22 \text{ V/\AA}$, beyond which the 90° AM configuration becomes energetically favored, identifying the critical switching field. At $0.3 \text{ V/\AA}$, the 90° configuration is clearly stabilized with a significant energy advantage.

To confirm that the altermagnetic character is preserved under the field, we calculated the band structure of the 90° configuration at $0.3 \text{ V/\AA}$ (**Figure 5d**). The spin splitting along the M- Γ -M' path remains around 82 meV (close to the zero-field value), confirming that the electric field locks the ligand orientation without perturbing the AM electronic structure. When the field is removed, the constrained-rotation energy landscape (**Figure 2e**) decreases monotonically from the 90° configuration toward the planar ground state, so the system relaxes back to the 0° AFM state, establishing full reversibility of the switching.

These results demonstrate that the electric field serves as an effective external switch for the altermagnetic state. It synchronizes the ligand orientations into the 90° configuration and locks them in place, overcoming the thermal disorder that prevents long-range AM order. The switching is reversible, field-tunable, and does not compromise the large AM spin splitting.

We note two limitations of the present study. First, the energy difference between the AFM and AM states at 0° is small ($\sim 3 \text{ meV}$), so the magnetic ground-state ordering may be sensitive to the exchange-correlation functional; benchmarking with hybrid functionals is left for future work. Second, the collective rotation pathway was sampled through constrained relaxations rather than nudged elastic band calculations; the full minimum-energy pathway will be reported elsewhere. The thermal stability discussed

here is supported by AIMD at 300 K, consistent with the practice of recent theoretical studies of MOF altermagnets.^{14,40}

3. Conclusions

We have demonstrated a reversible electric-field switch between antiferromagnetic (AFM) and altermagnetic (AM) states in a d^0 -metal-organic framework, where the magnetic response originates entirely from the ligand p orbitals. The switching is driven by a purely geometric operation—ligand rotation—which reorients the π orbitals from out-of-plane (p_z -like) to in-plane (p_x/p_y -like), thereby reversing the J_1/J_2 superexchange competition and toggling the magnetic ground state. The resulting AM phase exhibits a large spin splitting of approximately 84 meV, among the largest reported for MOF-based altermagnets and, notably, arising entirely from ligand p orbitals. Unlike previous approaches that rely on static design of MOF magnets, our work introduces a dynamic control paradigm for p -electron altermagnetism. The ligand rotation is enabled by the structural flexibility inherent to MOFs, and the metastable 90° AM state is locked by a moderate electric field of ~ 0.22 V/Å. Thermal fluctuations at 300 K can only produce transient local tilts with sub-picosecond lifetimes; the electric field serves as the essential agent to synchronize and stabilize long-range AM order. As the constrained-rotation energy landscape decreases monotonically back to the planar AFM ground state, the system is expected to relax back spontaneously upon field removal, completing a reversible switching cycle, while the AM spin splitting remains intact under the field.

By leveraging d^0 metal centers, we isolate magnetism to the ligand sublattice, establishing a clean platform for p -electron altermagnetism. The observed orbital-selective superexchange switching provides a general mechanism for designing responsive magnetic materials in MOFs, where ligand geometry serves as a direct control knob for magnetic order. More broadly, this strategy can be extended to other polar substituents, highlighting the chemical flexibility of the approach for experimental realization. Our findings open a new avenue for organic spintronics, where the structural flexibility and chemical tunability of MOFs can be exploited for on-demand magnetic switching. The combination of large spin splitting, a moderate

switching field, and reversible operation positions this platform as a promising candidate for future low-power, reconfigurable spintronic devices.

Supporting Information

Calculation details; torsion-modulated four magnetic orders; C/N sites; flipped-linker band structures; AIMD energy trajectories of the fluorinated Ca(TPN)₂ monolayer; band structures and PDOS at 0°/90° conformations; spin charge densities of fluorinated monolayers; structural, energetic and magnetic exchange evolutions of fluorinated and methylated Ca(TPN)₂ monolayers; lattice parameters; relative energies of three magnetic states; J_1 and J_2 exchange coupling constants.

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Notes

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Figures

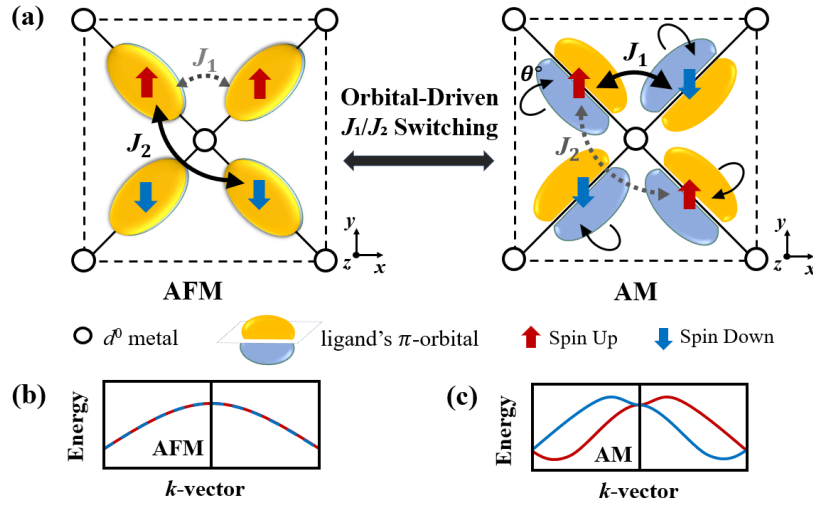


Figure 1. Design concept of the 2D d^0 -MOF. (a) Schematic of the two limiting ligand configurations: the planar configuration (0°, left) with AFM spin ordering and the perpendicular configuration (90°, right) with AM spin ordering. The nearest-neighbor J_1 (90° ligand-metal-ligand) and next-nearest-neighbor J_2 (180° ligand-metal-ligand) pathways are marked; bold black lines highlight the dominant exchange channel in each case (J_2 for 0°, J_1 for 90°). (b) Schematic band structure of the AFM state showing spin-degenerate bands. (c) Schematic band structure of the AM state showing momentum-dependent spin splitting.

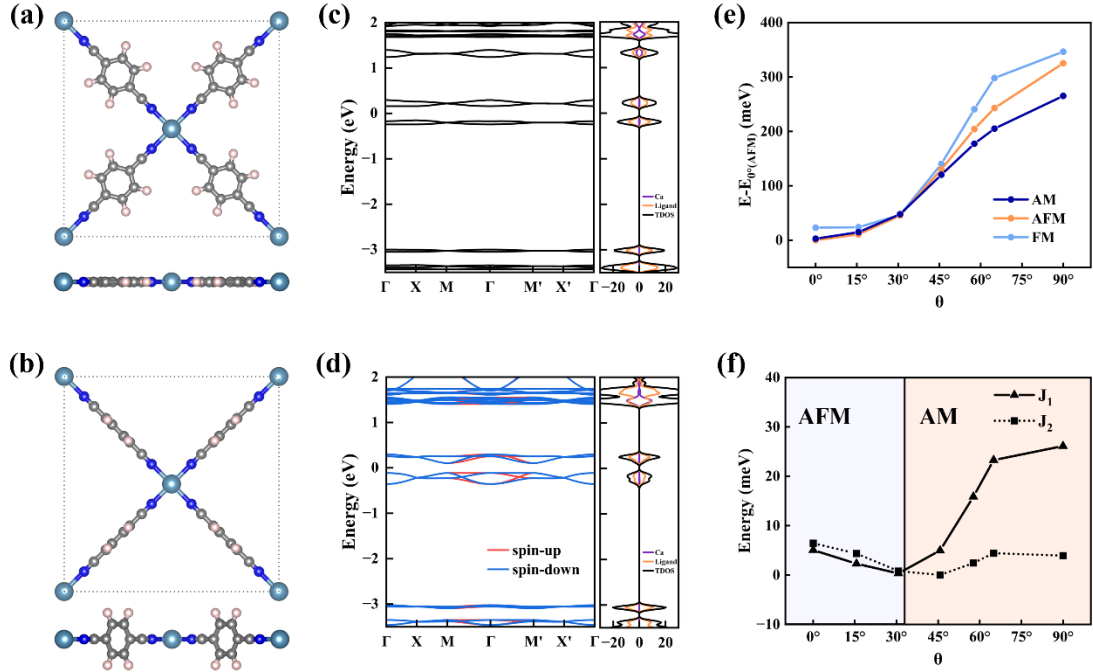


Figure 2. Structure and electronic properties of the Ca(TPN)₂ monolayer. (a) Planar configuration (0°). (b) Perpendicular configuration (90°), using a $\sqrt{2} \times \sqrt{2}$ supercell. Top and side views are shown; teal, blue, gray, and white spheres represent Ca, N, C, and H atoms, respectively. (c) Band structure and PDOS of the 0° configuration in the AFM state. (d) Band structure and PDOS of the 90° configuration in the AM state. (e) Relative energies of AFM, AM, and FM states as a function of ligand tilt angle, with the 0° AFM state set as zero. (f) Exchange parameters J_1 and J_2 as a function of ligand tilt angle.

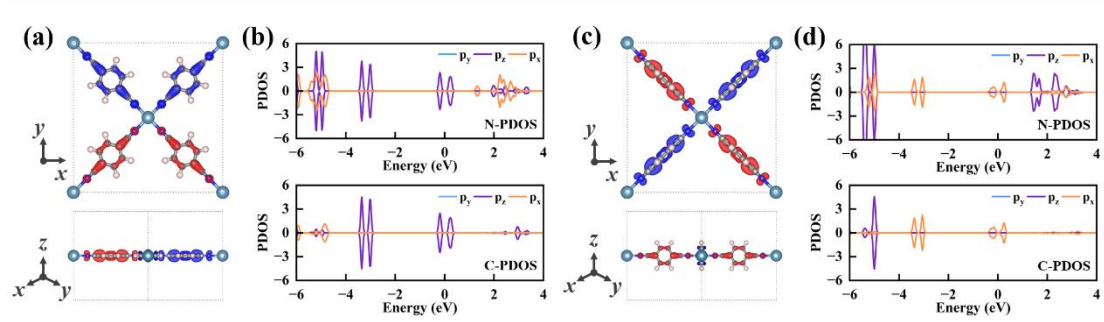


Figure 3. Spin charge density and projected density of states (PDOS) of the $\text{Ca}(\text{TPN})_2$ monolayer. (a,c) Top and side views of spin charge density for the 0° and 90° configurations, respectively. Red and blue isosurfaces denote spin-up and spin-down charge densities (isovalue = $0.004 \text{ e}/\text{bohr}^3$). (b,d) PDOS of the N and C atoms that contribute most to the magnetic moment in the 0° and 90° configurations, respectively.

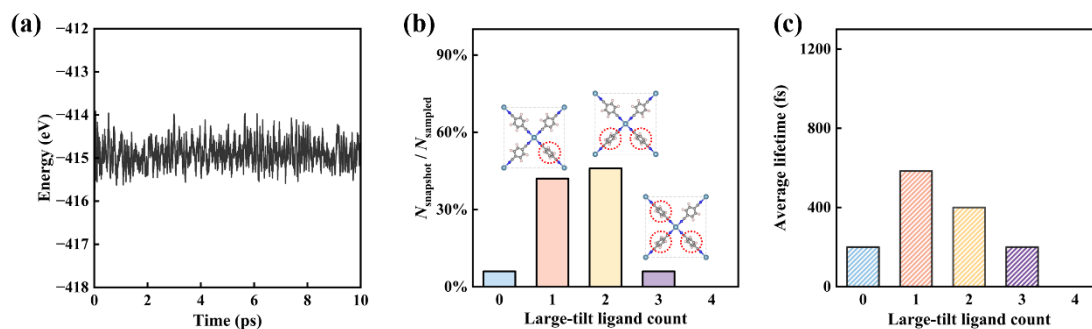


Figure 4. AIMD simulation of the $\text{Ca}(\text{TPN})_2$ monolayer at 300 K. (a) Total energy evolution over 10 *ps*. (b) Distributions of snapshots showing no flip, single-ligand flip, dual-ligand flip, and three- or four-ligand flip events (bars), with representative structures of each event type shown in the insets. (c) Average lifetimes of the corresponding flipped configurations extracted from the trajectory.

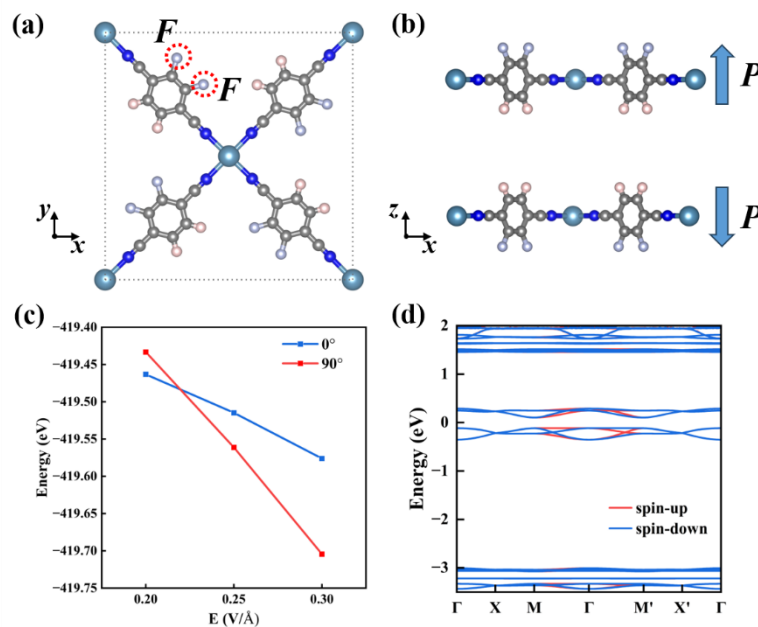


Figure 5. Electric-field switching in the fluorinated $\text{Ca}(\text{TPN})_2$ monolayer. (a) Chemical structure of the half-fluorinated TPN ligand, with F atoms highlighted. (b) Side view of the monolayer showing the net dipole moment direction (P) of the 2D MOF arising from the oriented fluorinated ligands. (c) Energy difference between the 0° and 90° configurations as a function of applied electric field (0.2–0.3 V/Å). (d) Band structure of the 90° configuration under a field of 0.3 V/Å, showing that the large AM splitting is preserved.

Electric-Field-Switchable Altermagnetism *via* Ligand Rotation in a d^0 Metal–Organic Framework

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Calculation details

First-principles density functional theory (DFT) calculations were performed using the Vienna Ab initio Simulation Package (VASP).^{1,2} The exchange-correlation interactions were described by the Perdew–Burke–Ernzerhof (PBE) generalized gradient approximation³, while Grimme’s DFT-D3 dispersion correction was included to capture van der Waals forces⁴. A vacuum region of ~ 15 Å along the z-axis was introduced to suppress spurious interlayer coupling between periodic images. The plane-wave cutoff energy was fixed at 500 eV. Structural relaxation proceeded until the total energy converged to 10^{-5} eV and the residual atomic forces fell below 0.01 eV/Å. A Monkhorst–Pack k-point mesh of $3 \times 3 \times 1$ was used for both geometry optimization and electronic structure calculations.⁵ Ab initio molecular dynamics (AIMD) simulations in the canonical (NVT) ensemble at 300 K were implemented to evaluate the thermal stability of the monolayer. Temperature regulation was realized via the Nosé–Hoover thermostat⁶⁻⁹, and the same $3 \times 3 \times 1$ k-grid was adopted for dynamic simulations. All output files were post-processed with the VASPKIT.¹⁰ The Monte Carlo simulations employed a $20 \times 20 \times 1$ supercell with periodic boundary conditions, as implemented in the SEU-mtc package¹¹. A temperature step of 1 K was used in the range of 1 K to 100 K.

Given that all magnetic interactions are exclusively mediated by superexchange through the TPN ligands, the Heisenberg spin Hamiltonian can be expressed as

$$H = J_1 \sum_{\langle i,j \rangle_1} S_i \cdot S_j + J_2 \sum_{\langle i,j \rangle_2} S_i \cdot S_j \quad (\text{S1})$$

where J_1 , J_2 and S denote the exchange coupling constants and the unit spin vector aligned along the c-axis, respectively. We identify two distinct superexchange pathways in this lattice: the nearest-neighbor pathway along the square edge defined as J_1 , corresponding to the 90° ligand–metal–ligand bridging geometry; and the next-nearest-neighbor diagonal pathway denoted as J_2 , which arises from the 180° ligand–metal–ligand superexchange coupling. The spin quantum number for each ligand-based magnetic unit is set to $S = 1/2$.

By incorporating both the 90° J_1 edge-path and 180° J_2 diagonal-path ligand–metal–ligand superexchange interactions, the total energies of ferromagnetic (FM), antiferromagnetic (AM), and antiferromagnetic (AFM) spin configurations can be formulated as follows:

$$E_{AM} = E_0 - 8J_1S^2 + 4J_2S^2 \quad (\text{S2})$$

$$E_{FM} = E_0 + 8J_1S^2 + 4J_2S^2 \quad (S3)$$

$$E_{AFM} = E_0 - 4J_2S^2 \quad (S4)$$

Accordingly, the exchange parameters J_1 and J_2 can be derived from the energy differences of distinct magnetic states:

$$J_1 = \frac{E_{FM} - E_{AM}}{16S^2} \quad (S5)$$

$$J_2 = \frac{8J_1S^2 - (E_{AFM} - E_{AM})}{8S^2} \quad (S6)$$

Herein, positive and negative signs correspond to parallel and antiparallel spin arrangements, respectively.

Supporting Figures and Tables

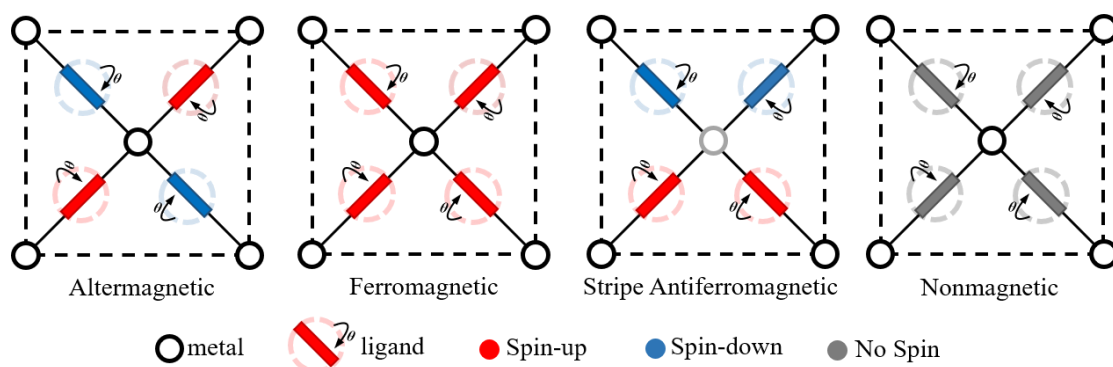


Figure S1. Four types of ligand-scale magnetic ordered structures modulated by ligand torsion angles (the dashed circles at the bottom correspond to the planar conformation at 0° , and the rectangular regions correspond to the twisted conformation at 90°): altermagnetic, ferromagnetic, stripe antiferromagnetic and nonmagnetic configurations. Red denotes spin-up ligands, blue denotes spin-down ligands, gray denotes spinless ligands, and hollow circles represent metal sites in the lattice.

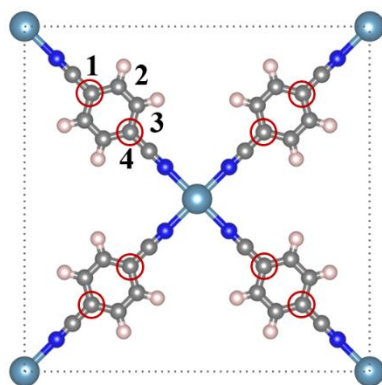


Figure S2. Selection of C and N Sites for PDOS Analysis. The eight circled red C atoms at the 1,4 positions and their eight corresponding N atoms are adopted to compute the partial density of states (PDOS) for the 0° and 90° conformations of pristine, fluorinated, and methylated $\text{Ca}(\text{tpn})_2$. Teal, blue, gray, and white spheres represent Ca, N, C, and H atoms, respectively.

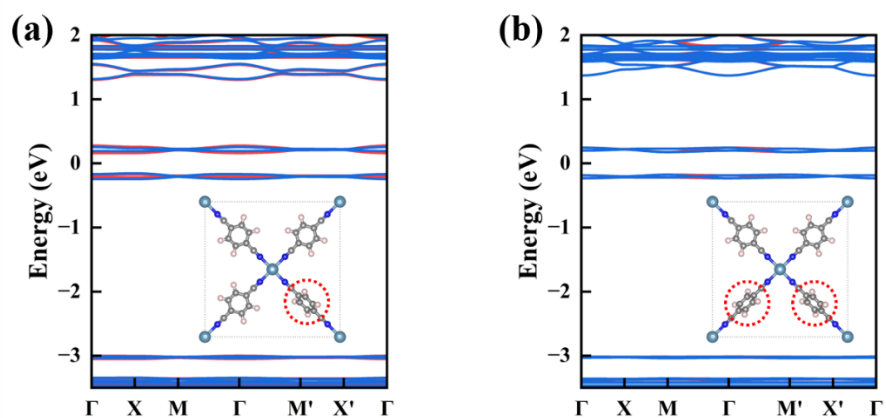


Figure S3. Band structures of thermally disordered ligand-flip snapshots: (a) single tilted ligand, (b) dual-tilt ligands. Spin splitting is strongly suppressed (~ 15 meV in Figure b) relative to fully ordered altermagnetic state.

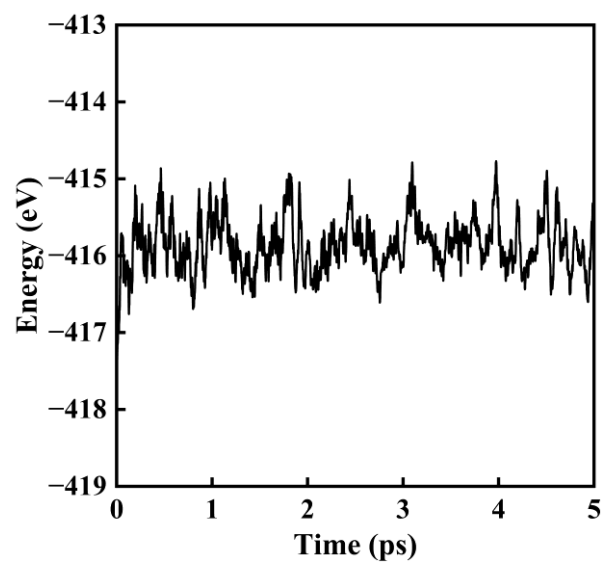


Figure S4. AIMD simulation of the fluorinated $\text{Ca}(\text{TPN})_2$ monolayer at 300 K. Total energy evolution over 5 *ps*.

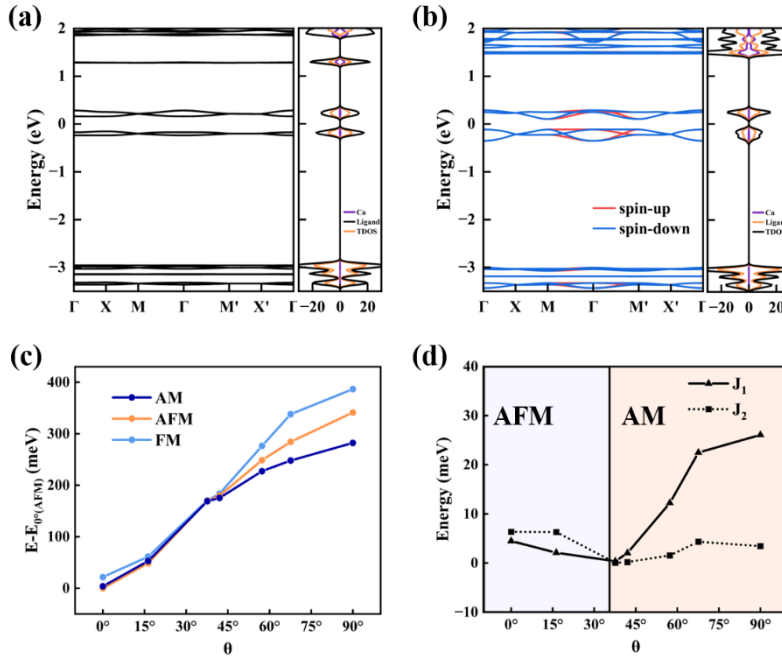


Figure S5. Electronic properties of the fluorinated $\text{Ca}(\text{TPN})_2$ monolayer. (a) Band structure and PDOS of the 0° configuration in the AFM state. (b) Band structure and PDOS of the 90° configuration in the AM state. (c) Relative energies of AFM, AM, and FM states as a function of ligand tilt angle, with the 0° AFM state set as zero. (d) Exchange parameters J_1 and J_2 as a function of ligand tilt angle.

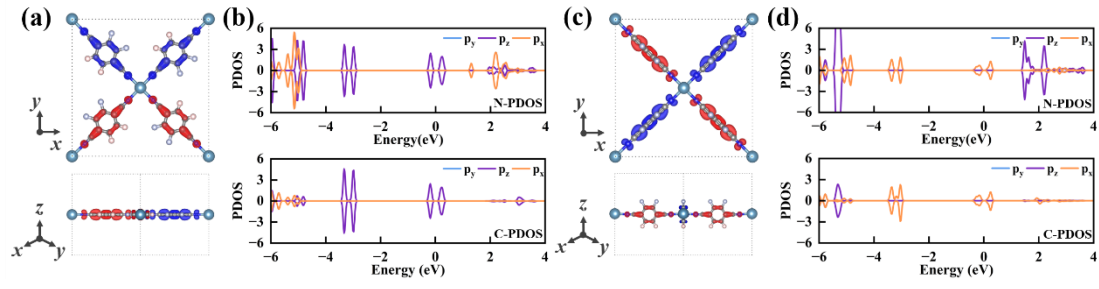


Figure S6. Spin charge density and projected density of states (PDOS) of the fluorinated $\text{Ca}(\text{TPN})_2$ monolayer. (a,c) Top and side views of spin charge density for the 0° and 90° configurations, respectively. Red and blue isosurfaces denote spin-up and spin-down charge densities (isovalue = 0.004 e/bohr^3). (b,d) PDOS of the N and C atoms that contribute most to the magnetic moment in the 0° and 90° configurations, respectively.

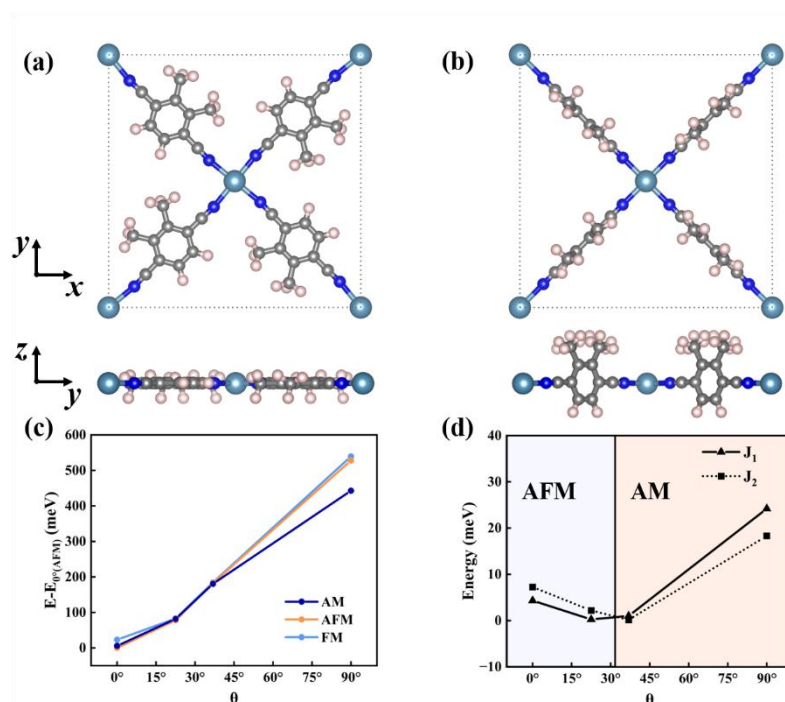


Figure S7. Structural, energetic, and magnetic exchange evolution of methylated $\text{Ca}(\text{TPN})_2$ monolayer. (a) Planar configuration (0°). (b) Perpendicular configuration (90°), using a $\sqrt{2} \times \sqrt{2}$ supercell. Top and side views are shown; teal, blue, gray, and white spheres represent Ca, N, C, and H atoms, respectively. (c) Relative energies of AFM, AM, and FM states as a function of ligand tilt angle, with the 0° AFM state set as zero. (d) Exchange parameters J_1 and J_2 as a function of ligand tilt angle.

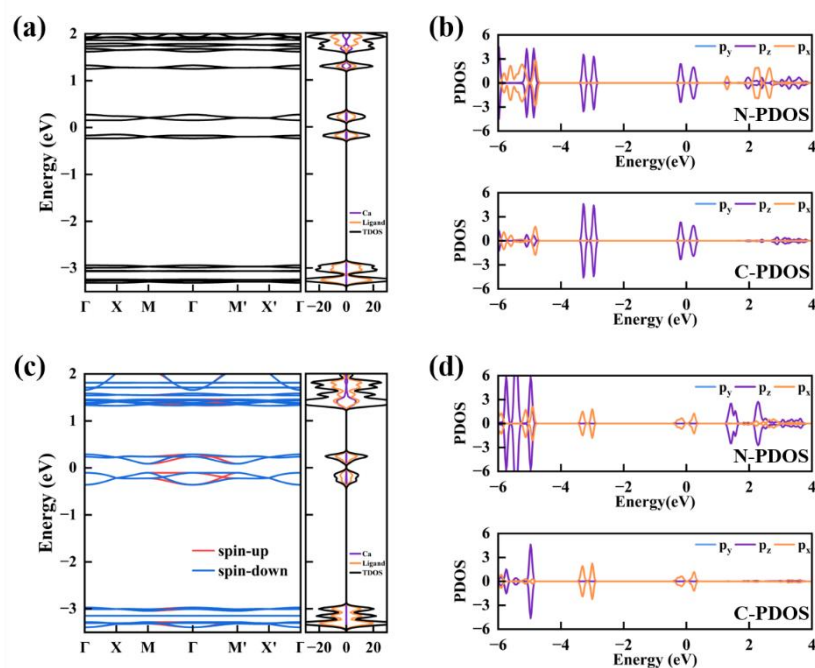


Figure S8. Electronic properties and projected density of states (PDOS) of methylated $\text{Ca}(\text{TPN})_2$ monolayer. (a,c) Band structure and PDOS of the 0° and 90° configuration, respectively. (b,d) PDOS of the N and C atoms that contribute most to the magnetic moment in the 0° and 90° configurations, respectively.

Table S1. Lattice parameters (L for length and α for angle), local magnetic moments (μ_M) on metal cations and local magnetic moments (μ_L) on ligand anions of $\text{Ca}(\text{TPN})_2$ and fluorinated $\text{Ca}(\text{TPN})_2$.

	$\text{Ca}(\text{TPN})_2(0^\circ)$	$\text{Ca}(\text{TPN})_2(90^\circ)$	$\text{Ca}(\text{Hf-TPN})_2(0^\circ)$	$\text{Ca}(\text{Hf-TPN})_2(90^\circ)$
L (Å)	17.92	17.90	17.93	17.91
α (°)	90	90	90	90
μ_M (μ_B)	0	0	0	0
μ_L (μ_B)	0.50	0.43	0.50	0.43

Table S2. Energies (in meV per unit cell) of the altermagnetic (AM), ferromagnetic (FM), antiferromagnetic (AFM), and nonmagnetic (NM) states, and calculated nearest-neighbor (J_1) and next-nearest-neighbor (J_2) exchange parameters, for pristine hydrogen-based and fluorinated 90° Ca(TPN)₂ monolayers. All energies are referenced to the respective ground state.

	Ca(TPN) ₂ (0°)	Ca(TPN) ₂ (90°)	Ca(Hf-TPN) ₂ (0°)	Ca(Hf-TPN) ₂ (90°)
E_{AM}	2.78	0	3.74	0
E_{FM}	23.02	104.42	21.64	104.35
E_{AFM}	0	60.07	0	58.97
E_{NM}	245.02	103.53	246.36	103.56
J_1	5.06	26.11	4.48	26.09
J_2	6.45	-3.94	6.35	-3.40

Table S3. Calculated lattice parameters (L , in Å; α , in °), local magnetic moments on metal cations (μ_M) and ligand anions (μ_L), magnetic energies (E , in meV per unit cell) referenced to the ground state, and exchange coupling constants J_1 and J_2 for methylated Ca(TPN)₂.

	L (Å)	α (°)	μ_M (μ_B)	μ_L (μ_B)	E_{AM}	E_{FM}	E_{AFM}	J_1	J_2
0°	17.81	90	0	0.49	5.86	23.12	0	4.32	7.25
90°	17.87	90	0	0.42	0	96.74	85.19	24.19	-18.32

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